GS FOUNDATION 2024-25

GEOGRAPHY - 28

Resources of India - I

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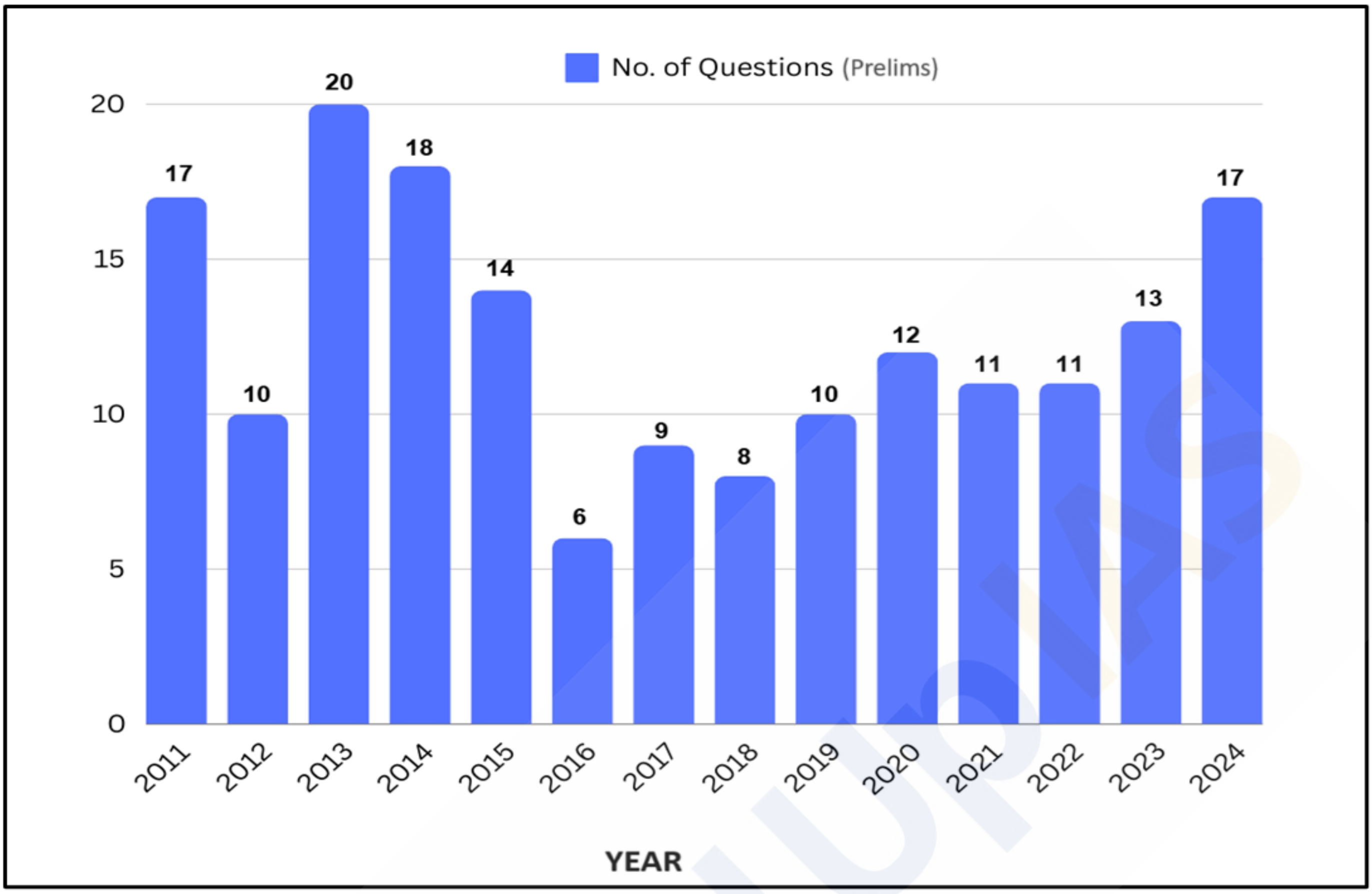

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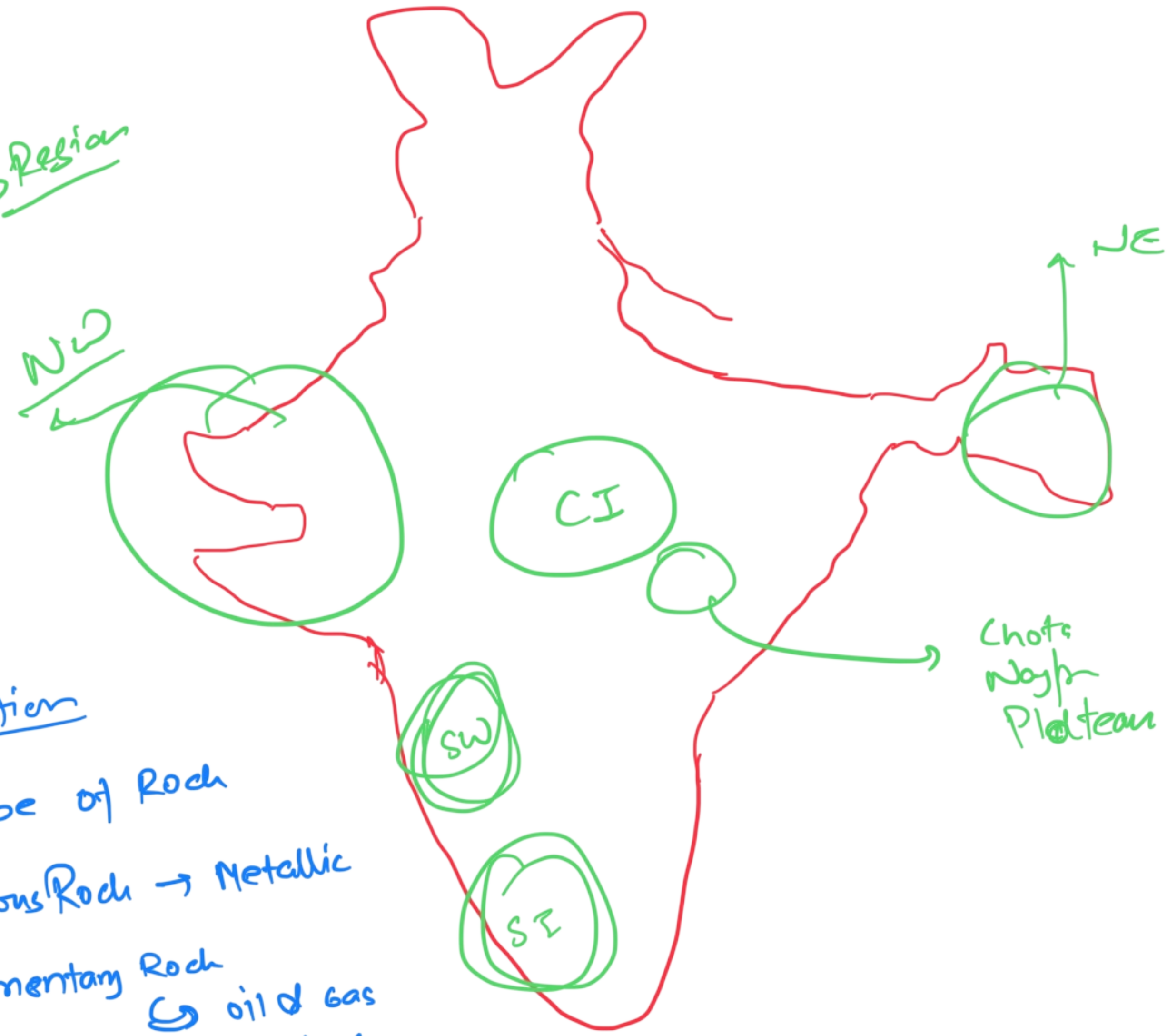
Mineral Resource belt of country



Distribution

(1) Type of Rock

Igneous Rock → Metallic

Sedimentary Rock
↳ oil & Gas
Shale Gas

(2) Biological Factor : Forest decomposition
Marine decomposition

(3) Climate → laterisation
↳ Al. accumation → Bauxite

(4) Tectonic factor → Rifting →

(5) Marine Transgression

Mineral Resources

Introduction:

The Earth is composed of various elements, existing in solid form on the outer layer and in hot, molten form in the interior. **About 98% of the Earth's crust is made up of eight primary elements - oxygen, silicon, aluminium, iron, calcium, sodium, potassium,** and **magnesium.** The remaining 2% consists of elements such as titanium, hydrogen, phosphorus, manganese, sulphur, carbon, nickel, and others.

The elements in the earth's crust are rarely found exclusively but are usually **combined with other elements to make various substances.** Though the lithosphere contains a limited number of elements, these elements combine in numerous ways, resulting in a wide variety of minerals. **Over 2,000 minerals have been identified in the Earth's crust,** but the **majority belong to six major rock-forming mineral groups.**

The primary source of all minerals is the hot magma within the Earth. As magma cools, mineral crystals form in a specific sequence, solidifying to create rocks. Minerals such as coal, petroleum, and natural gas are organic and found in solid, liquid, and gaseous forms, respectively.

Mineral

*A mineral is **a naturally occurring** substance that can be **either organic or inorganic.** It possesses an **orderly atomic structure,** along with a **specific chemical composition** and **distinct physical properties.** Typically, a mineral consists of **two or more elements.** However, there are instances where minerals consist of **single elements,** such as sulphur, copper, silver, gold, and graphite.*

Mineral Wealth of India:

India is endowed with a **rich variety of minerals**, thanks to its large size and diverse geological formations. It is estimated that **nearly 100 minerals** are produced or worked in India, with **approximately 30** being considered **significant.** Among these, several minerals may currently be of limited quantity but have the potential for significant development as industrial needs expand.

India possesses **abundant reserves** of **coal**, **iron**, and **mica**, alongside **adequate supplies** of **manganese ore**, **titanium**, and **aluminium**. The country also has essential raw materials for **refractories** and **limestone**. However, there are **notable deficiencies** in the availability of **copper**, **lead**, and **zinc** ores, while **workable deposits** of **tin** and **nickel** can be found.

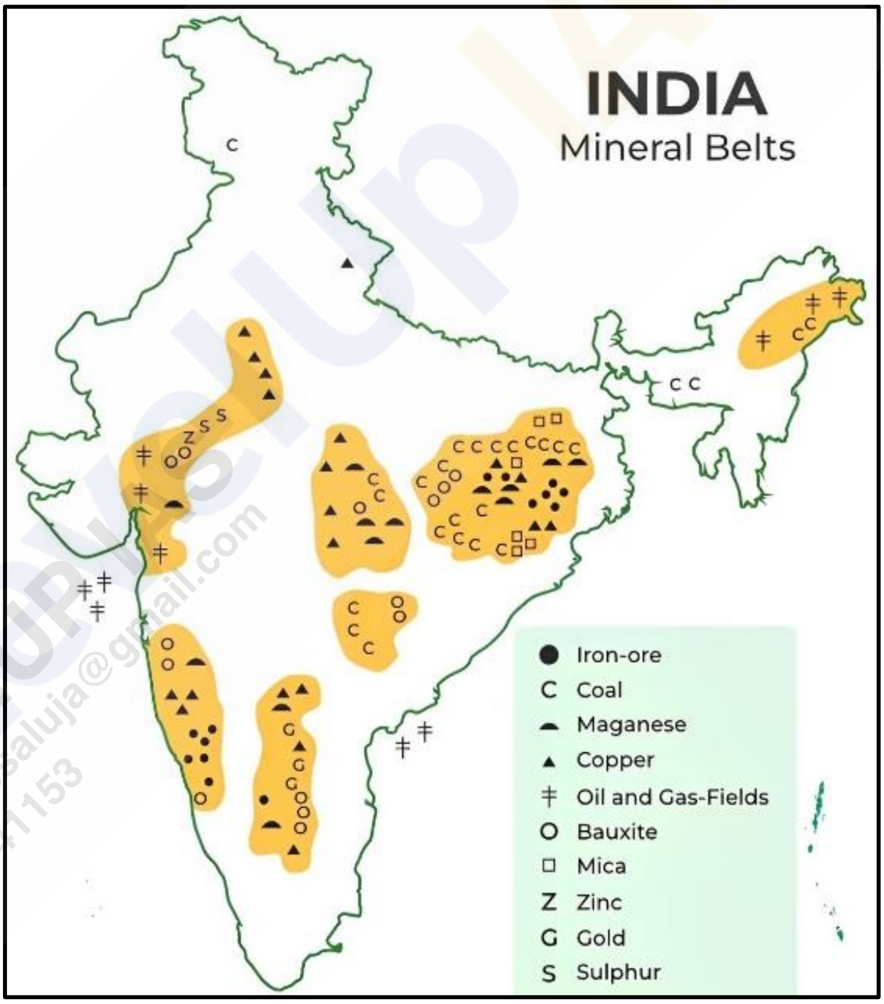
India generates substantial foreign exchange by exporting a diverse array of minerals, including **iron ore**, **titanium**, **manganese**, **bauxite**, **granite**, and many others. However, the country must rely on imports to satisfy its requirements for various minerals such as **copper**, **silver**, **nickel**, **cobalt**, **zinc**, **lead**, **tin**, **mercury**, **limestone**, **platinum**, **graphite**, and numerous other minerals. This dual aspect of mineral wealth highlights both the **economic potential** and the **challenges** faced by India in the management of its mineral resources.

Distribution of Minerals in India:

The **distribution of minerals in India** is characterized by a pronounced **unevenness,** where some regions are abundantly rich in mineral resources while others are almost devoid of them. This disparity can be attributed to various geographical and climatic factors.

The high rainfall areas, such as those in the **Western Ghats** and **Northeastern regions, lack soluble minerals** like **limestone, gypsum, and salts**. Similarly, the **northern plains of India,** covered with thick layers of alluvium, conceal bedrock and are **poor in mineral resources.** The **Himalayan region** possesses a variety of rocks, but its **complex geological structure and challenging terrain** make **mineral exploitation economically unfeasible.** The lack of substantial mineral resources in this region is compounded by difficulties in transportation, sparse population, and adverse climatic conditions. Conversely, India's most significant mineral deposits are found in the **old, crystalline rock structures** of the **plateau** and low hill regions of **peninsular India**, which are organized into well-defined mineral belts.

The North-Eastern Peninsular Belt:



- The North-Eastern Peninsular Belt includes the **Chhota Nagpur Plateau** and the **Odisha Plateau,** spanning the states of **Jharkhand, West Bengal, and Odisha**. This belt is the **richest mineral region in India,** hosting an extensive range of minerals. It is notable for containing substantial quantities of **coal, iron ore, manganese, mica, bauxite, copper, kyanite, chromite, beryl, and apatite**. The Chhota Nagpur Plateau, often referred to as the **mineral heartland of India,** holds a significant proportion of the country's mineral resources. According to initial studies, this region contains **100% of India's kyanite, 93% of iron ore, 84% of coal, 70% of chromite, 70% of mica, 50% of fire clay, 45% of asbestos, 45% of China clay, 20% of limestone**, and **10% of manganese**. Recent changes have, however, affected these proportions.

The Central Belt:

- The Central Belt extends over parts of **Chhattisgarh, Madhya Pradesh, Telangana, Andhra Pradesh**, and **Maharashtra**. This belt is the **second-largest mineral-rich area** in India, with substantial deposits of **manganese, bauxite, limestone, marble, coal, gems (such as panna), mica, iron ore**, and **graphite.** Despite its significant mineral wealth, a comprehensive geological survey is still needed to fully understand the extent of its resources.

The Southern Belt:

- The Southern Belt primarily covers the **Karnataka Plateau** and extends into the contiguous uplands of **Tamil Nadu**. While it shares similarities with the North-Eastern Peninsular Belt in terms of **ferrous minerals** and **bauxite**, it lacks significant coal deposits, except for **lignite at Neyveli.** Additionally, it does not have substantial mica or copper deposits, making its mineral diversity less pronounced compared to the North-Eastern Peninsular Belt.

The South-Western Belt:

- The South-Western Belt includes parts of **Western Karnataka** and **Goa**, where deposits of **iron ore, garnet, and clay** are found. This belt contributes to India's mineral resources but is less varied compared to other major belts.

The North-Western Belt

- The North-Western Belt stretches along the **Aravalis in Rajasthan** and adjacent parts of **Gujarat**. This belt has developed recently and shows promise for the mining of **non-ferrous metals** such as **copper, lead, and zinc**, as well as **uranium, mica, steatite, beryllium**, and **precious stones** like **aquamarine and emerald**. Gujarat, in particular, is emerging as a significant producer of **petroleum** and also produces **gypsum, manganese, salt, and bauxite**.

Other Mineral Deposits:

- Outside these main belts, mineral deposits are scattered in various other regions of India. For instance, **Assam** has reserves of **petroleum** and **lignite**, while the **Himalayan region** contains some deposits of **coal, bauxite, copper**, and **slate**. Additionally, the **Mumbai High** and the **Godavari Basin** are known for their reserves of **oil** and **natural gas**.
- With advancements in technology, the potential for **sea bed mineral exploitation** has increased. There are significant possibilities for obtaining large quantities of **oil, natural gas, uranium, manganese, copper, zinc**, and **lead** from the sea bed, marking an exciting frontier for future mineral exploration and extraction.

Types of Minerals:

Minerals can be broadly classified into **two categories - metallic and non-metallic minerals.** This classification is based on the presence or absence of metal content in the minerals.

Metallic Minerals:

- Metallic minerals are those that **contain metal elements.** These minerals are valuable due to their metallic content and are widely used in various industries, especially in construction, manufacturing, and technology. Some of the most important examples of metallic minerals include **iron ore**, **copper**, **manganese**, and **nickel**.
- **Metallic minerals** are further divided into **two subcategories:**
  - **Ferrous Minerals -** These minerals contain iron, making them crucial for industries like steel production. Common examples of ferrous minerals are **iron ore**, **manganese**, **chromite**, **pyrites**, **tungsten**, **nickel**, and **cobalt**. The presence of iron is what defines this group of minerals.
  - **Non-ferrous Minerals -** Unlike ferrous minerals, these do not contain iron. Non-ferrous minerals are essential in industries like electronics, aerospace, and packaging. Examples include **gold**, **silver**, **copper**, **lead**, **bauxite**, **tin**, and **magnesium**. Since they lack iron content, these minerals are resistant to rust and corrosion, making them useful for specific applications.

Non-metallic Minerals:

- Non-metallic minerals, as the name suggests, **do not contain metal elements.** These minerals are widely used in construction, agriculture, and manufacturing. Examples of non-metallic minerals include **limestone**, **nitrate**, **potash**, **dolomite**, **mica**, and **gypsum**. These minerals play an essential role in industries such as cement production, fertilizers, and building materials.
- Additionally, **coal** and **petroleum** are also classified as non-metallic minerals. While they do not fit the traditional definition of minerals, they are extremely valuable as **mineral fuels, providing energy** for transportation, heating, and electricity generation.

Metallic Minerals:

Metallic minerals play a significant role in India's mining sector and form the **backbone of the country's metallurgical industries.** Among the many metallic minerals, **iron ore** is one of the most important, serving as a crucial raw material for the **steel industry** and being essential for modern civilization. It is extracted from mines and is processed for various industrial purposes. Iron ore is found in different types, each varying in quality based on the percentage of pure iron they contain. The following are the **4 major types of iron ores found in India.**

- **Haematite:**
  - Haematite is considered the **best quality of iron ore,** with about **70% metallic content**. This ore is generally found as a **massive, hard, compact** form with a **reddish or coral red colour**. It occurs mainly in the **Dharwad** and **Cuddapah rock systems** of peninsular India. The **majority (over 80%)** of India's haematite reserves are located in the eastern states, such as **Odisha, Jharkhand, Chhattisgarh**, and **Andhra Pradesh**. In western India, haematite ores are found in **Karnataka, Maharashtra**, and **Goa**. This high-quality iron ore forms the bulk of India's iron ore production.
- **Magnetite:**
  - Magnetite, often called **'black ore'** due to its blackish appearance, is the **second-best form of iron ore,** with a **metallic content** ranging from **60% to 70%**. It is also found in the **Dharwad** and **Cuddapah rock systems** and is distinguished by its **magnetic properties**. This quality makes it easily identifiable as magnetite ore. **Karnataka, Andhra Pradesh, Rajasthan, Tamil Nadu**, and **Kerala** hold significant magnetite reserves.
- **Limonite:**
  - Limonite is considered an **inferior ore** compared to haematite and magnetite, with a **metallic content** ranging from **40% to 60%**. This ore has a **yellowish colour** and occurs in specific geological settings like the **Damuda series** in the **Raniganj coal field**. Limonite ores are also found in **Garhwal** (Uttarakhand), **Mirzapur** (Uttar Pradesh), and the **Kangra Valley** (Himachal Pradesh). Despite being lower in quality, limonite has the advantage of **easy and cost-effective mining**, making it economically feasible to extract in certain regions.
- **Siderite:**
  - Siderite, also referred to as **'iron carbonate'**, is an iron ore of **inferior quality** with a **metallic content** of **less than 40%**. This ore contains **several impurities** and, due to its poor iron content, is generally **not economically viable for large-scale mining.** However, siderite has a **self-fluxing property** due to the presence of **lime**, which can be beneficial during the smelting process.

Iron Ore:

- India is rich in iron ore reserves, with **hematite** and **magnetite** being the **two most important iron ores.** These ores form the backbone of India's iron and steel industry, which plays a crucial role in the country's economic growth. The iron ore reserves are classified based on the **United National Framework Classification (UNFC) of 2010**, which provides an **estimate of the total reserves and distribution** across various regions.

Hematite Ore Reserves:

Eastern Region → Odissa. Jharkhand, chattisgarh

- According to the **UNFC,** the total reserves of **hematite ore** in India are estimated at **17,882 million tonnes**. Hematite is primarily found in the **eastern region of the country.** The major sources of hematite reserves are concentrated in the following states:

- **Odisha -** Odisha holds the **largest share,** with reserves of **5,930 million tonnes**, accounting for **33%** of the total hematite reserves in the country.
- **Jharkhand -** The **second-largest** source, with **4,597 million tonnes** of hematite reserves, contributing **26%**.
- **Chhattisgarh -** Chhattisgarh holds **3,292 million tonnes**, which is about **18%** of India's hematite reserves.

- The remaining hematite reserves are distributed across states such as **Andhra Pradesh, Assam, Bihar, Maharashtra, Madhya Pradesh, Rajasthan,** and **Uttar Pradesh**.

Magnetite Ore Reserves:

- India's total **magnetite ore** reserves are estimated at **10,644 million tonnes**. Unlike hematite, magnetite reserves are concentrated in a few states, with **97%** of the total reserves located in just four states:
  - **Karnataka -** Karnataka is the leading state, with **7,802 million tonnes**, accounting for **73%** of the total magnetite reserves in India.
  - **Andhra Pradesh -** Andhra Pradesh follows with **1,464 million tonnes**, representing **14%** of the reserves.
  - **Rajasthan -** Rajasthan holds **527 million tonnes** of magnetite reserves, contributing **5%**.
  - **Tamil Nadu -** Tamil Nadu accounts for **4.9%** of the total magnetite reserves.
- The remaining **3.1%** of magnetite reserves are found in other states, including **Assam, Bihar, Goa, Jharkhand, Kerala, Maharashtra, Meghalaya, and Nagaland**.

Production and Distribution of Iron Ore in India:

- Iron ore is a critical resource for India's industrial sector, with about **95% of the country's total reserves** concentrated in a **5 key states** which include **Odisha, Jharkhand, Chhattisgarh, Karnataka** and **Goa,** reflecting a highly **uneven distribution** of resources across the country.

Production of Iron Ore in India

- **Odisha - 59.62%** (Leading Producer)
- **Chhattisgarh - 14.11%**
- **Karnataka - 12.76%**
- **Jharkhand - 10.93%**
- **Others - 2.56%**

Odisha:

- Odisha holds the **largest share of iron ore production,** contributing approximately **59%.** The state's major iron ore deposits are located at **Mayurbhanj (Badampahar), Banspani,** and **Toda in Kendujhar (Keonjhar)**, with significant reserves also in **Tomka Range in Cuttack, Kandadhar Pahar in Sundargarh, Sambalpur**, and **Hirapur Hills of Koraput district.**

Chhattisgarh:

- This state accounts for about **18%** of India's iron ore reserves. The principal iron ore mining regions are **Bailadila mine** (Asia's largest mechanized mine) **in the Bastar District** and **Dalli-Rajhara mine in the Durg district**. The iron ore found here includes both **haematite and magnetite** types, with metal content ranging from **60 to 70%**. The **Dalli-Rajhara** range in Durg, with reserves of 120 million tonnes, supplies iron ore to the **Bhilai Steel Plant**. Additionally, Chhattisgarh's **slurry pipeline** from **Bailadila to Vizag** has reduced the pressure on road transport, with high-grade ores exported to Japan.

Karnataka:

- Karnataka is a significant producer of iron ore, with **high-grade deposits primarily of haematite and magnetite.** While deposits are scattered across the state, the richest reserves are located in **Kemmangundi** in the **Baba Budan hills** of **Chikmagalur** district and **Sandur** and **Hospet** in **Bellary** district. The ore in these regions is predominantly high-grade **haematite** and **magnetite**. Other important producing districts include **Chitradurga, Uttar Kannada, Shimoga, Dharwar,** and **Tumkur**.

Jharkhand:

- **Jharkhand** accounts for **25%** of India's iron ore reserves and contributes over **11%** to national production. The **Singhbhum district**, where iron ore mining began in 1904, is known for its high-quality ore, capable of lasting hundreds of years. The major deposits are spread across a **50 km belt** from Gua to Bonai in Odisha. Key mining areas include **Budhu Buru, Kotamati Buru, Rajori Buru**, and the renowned **Noamandi mines**. Magnetite ores also occur in **Palamu** district and smaller quantities are found in **Santhal Parganas, Hazaribagh, Dhanbad,** and **Ranchi** districts.

Goa:

- Iron ore production in Goa began relatively late but experienced rapid growth, making Goa an important producer of iron ore in India. The **Geological Survey of India** identified **34 iron-bearing reserves** in 1975, estimating total deposits at 390 million tonnes. Significant deposits occur in **North Goa** (Pirna-Adolpale-Asnora, Sirigao-Bicholim-Daldal), **Central Goa** (Tolsia-Dongarvado-Sanvordem), and **South Goa** (Borgadongar, Netarlim, Rivona-Solomba). Most of Goa's ores are of low-grade **limonite** and **siderite**, with mechanized, open-cast mining ensuring efficient exploitation. Goa's iron ore is primarily exported to **Japan**, facilitated by the **Marmagao port**.

Other Producing States:

- **Andhra Pradesh** (Kurnool, Guntur, Cuddapah, Ananthapur, Nellore), **Maharashtra** (Chandrapur, Ratnagiri, Sindhudurg), **Madhya Pradesh** (Jabalpur, Gwalior), **Tamil Nadu** (Salem, Tiruchirapalli, Coimbatore, Madurai, Nellai Kattabomman), **Rajasthan** (Jaipur, Udaipur, Alwar, Sikar, Bundi, Bhilwara), **Uttar Pradesh** (Mirzapur), **Uttarakhand** (Garhwal, Almora, Nainital), **Himachal Pradesh** (Kangra, Mandi), **Haryana** (Mahendragarh), **West Bengal (**Burdwan, Birbhum, Darjeeling), **Jammu and Kashmir** (Udhampur, Jammu), **Gujarat** (Bhavnagar, Junagadh, Vadodara) and **Kerala** (Kozhikode).

Export of Iron Ore from India:

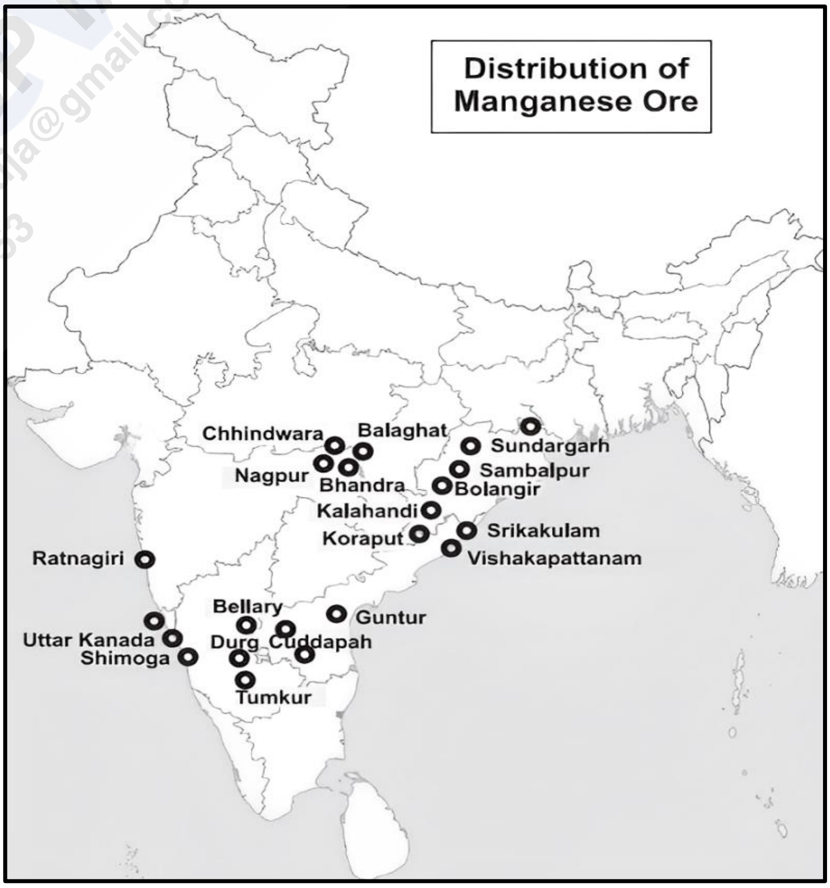
- **India** ranks as the **fifth-largest exporter of iron ore** globally, contributing significantly to the international iron ore market. A substantial portion, approximately **50 to 60%** of India's total iron ore production, is exported to various countries. The primary destinations for Indian iron ore exports include **Japan, South Korea, and several European nations**. In recent years, there has also been an increasing demand from the **Gulf countries**, expanding India's export base.
- Among all the countries, **Japan remains the largest importer** of Indian iron ore, accounting for roughly **three-fourths of India's total iron ore exports**. This long-standing trade relationship with Japan highlights the strategic importance of Indian iron ore in meeting the demands of Japan's steel industry. Other significant buyers include **South Korea** and **European countries**, which have continued to depend on India for a steady supply of high-quality iron ore.
- India's iron ore exports are facilitated through several key ports. The **Vishakhapatnam** port on the eastern coast is one of the largest hubs for handling iron ore shipments. Other major ports include **Paradip** on the Bay of Bengal, **Marmagao** on the western coast, and **Mangalore** in the southern region of India. These ports play a crucial role in connecting India's mining output with international markets, ensuring the smooth export of iron ore to various global destinations.

Manganese:

- **India** holds a significant position in the global manganese market, being the **second largest reserve holder of manganese** after **Zimbabwe** and the **fifth largest producer** following **Brazil, Gabon, South Africa,** and **Australia.** Manganese is a vital industrial metal, primarily used in the **manufacturing of iron and steel**. Its utility extends to various industries, including the production of **bleaching powder, insecticides, pesticides, paints, dry batteries**, and even **photography**.

Production and Distribution of Manganese in India:

- India's manganese reserves are **widely distributed** across several states. The leading states contributing to manganese production include **Odisha, Maharashtra, Madhya Pradesh, Karnataka,** and **Andhra Pradesh,** each with significant deposits.



Odisha:

- This state is the **largest producer of manganese** in India, contributing to over **37%** of the total production. The rich **Gondite deposits** in **Sundargarh** and **khondolite and kodurite deposits** in **Kalahandi and Koraput** are notable for their high-quality manganese. Additionally, production also takes place in the **Bolangir** and **Sambalpur** districts.

Maharashtra:

- Ranking second, Maharashtra contributes **23%** of the country's manganese production. The key manganese-producing districts include **Bhandara, Nagpur, and Ratnagiri**, with the ore from Ratnagiri being particularly renowned for its **superior quality**.

Madhya Pradesh:

- Around **20%** of India's manganese comes from Madhya Pradesh. The districts of **Balaghat** and **Chhindwara** are the principal regions where manganese mining takes place.

Karnataka:

- Accounting for **13%** of India's total production, Karnataka's manganese deposits are found in **North Kannada, Shimoga, Bellary, Chitradurga**, and **Tumkur**.

Andhra Pradesh:

- This state contributes **4.5%** of India's manganese production, with the primary deposits located in **Srikakulam** and **Vishakhapatnam** districts. Additionally, smaller deposits are found in **Cuddapah, Guntur,** and **Vijayanagram**.

Other States:

- Manganese is also mined in smaller quantities in states like **Goa**, **Gujarat** (notably in **Panchmahal and Vadodara**), **Rajasthan** (in **Banswara and Udaipur**), and **Jharkhand** (in **Dhanbad and Singhbhum**).

Export of Manganese from India:

- Approximately **80% of India's manganese production is consumed domestically,** predominantly by the **iron and steel industries**. The remaining production is exported to various countries, including **the USA, UK, Germany, France, Italy, Netherlands, Belgium, Czech Republic, Slovakia,** and several **East European nations**, such as **Ukraine**.
- However, **exports have been decreasing** due to the **rising domestic demand**. Despite this, **Japan remains the largest buyer** of Indian manganese, followed by **the USA, UK, Germany, France, and Norway**.

Chromite:

- Chromite is the only significant ore mineral of **chromium**, an essential element used in the production of various **alloys, most notably steel.** The mineral is an **oxide of iron and chromium**, with a theoretical composition of **32% iron** and **68% chromium**. However, the presence of impurities like **alumina, iron oxide, magnesium oxide, lime**, and **silica** can reduce the overall chromium content in the ore, affecting its quality.

Production and Distribution of Chromite in India:

- India is home to several major **chromite deposits**, with the state of **Odisha** leading in production. The primary regions where chromite is found are:

Odisha:

- **Odisha** is the **leading producer** of Chromite in India. The **Sukinda ultrabasic belt** in the districts of **Cuttack, Dhenkanal,** and **Keonjhar** is the largest chromite-producing region in the country. Odisha accounts for **93% of India's total chromite production**, making it the most significant contributor to the national output.

Karnataka:

- Chromite deposits are also found in Karnataka, specifically in the **Nuggchalli belt**, which includes areas such as **Byrapur, Chikonhalli, Pensamudra,** and **Bhaktarahalli** in **Hassan district.**

Maharashtra:

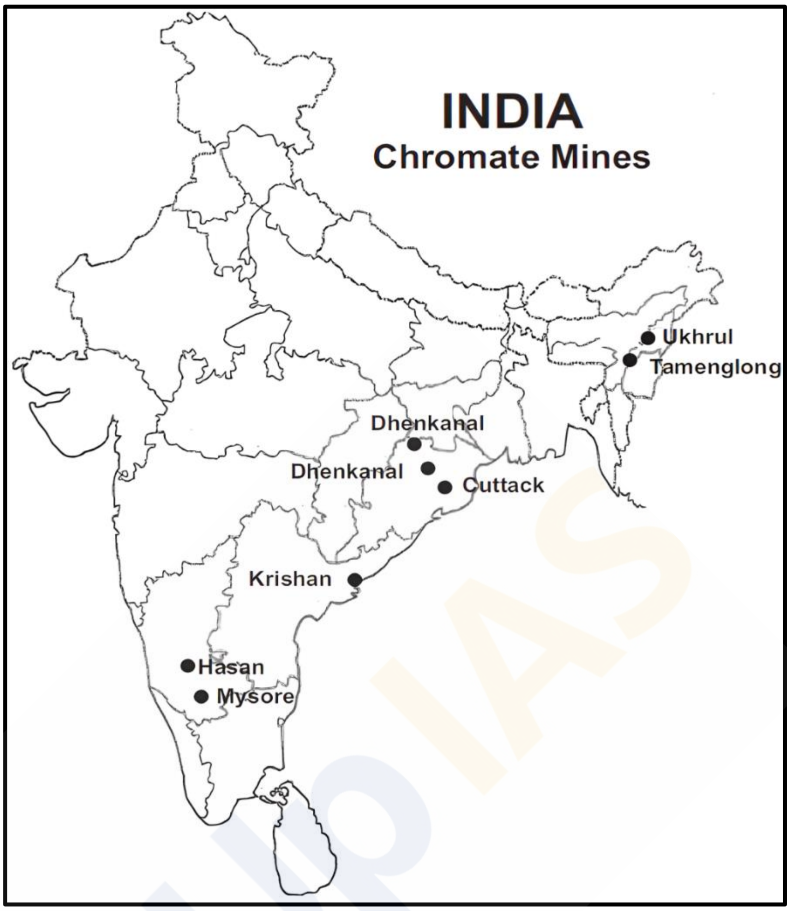
- In Maharashtra, chromite deposits are located in the **Kankauli** and **Vagda** areas of **Ratnagiri district.**

Jharkhand:

- Jharkhand hosts chromite in the **hills of Rorburu, Kiriburu,** and **Chittangburu** in the **Singhbhum district.**

Tamil Nadu:

- In **Tamil Nadu**, the significant chromite deposit is found in **Sittampudi** in **Salem district.**



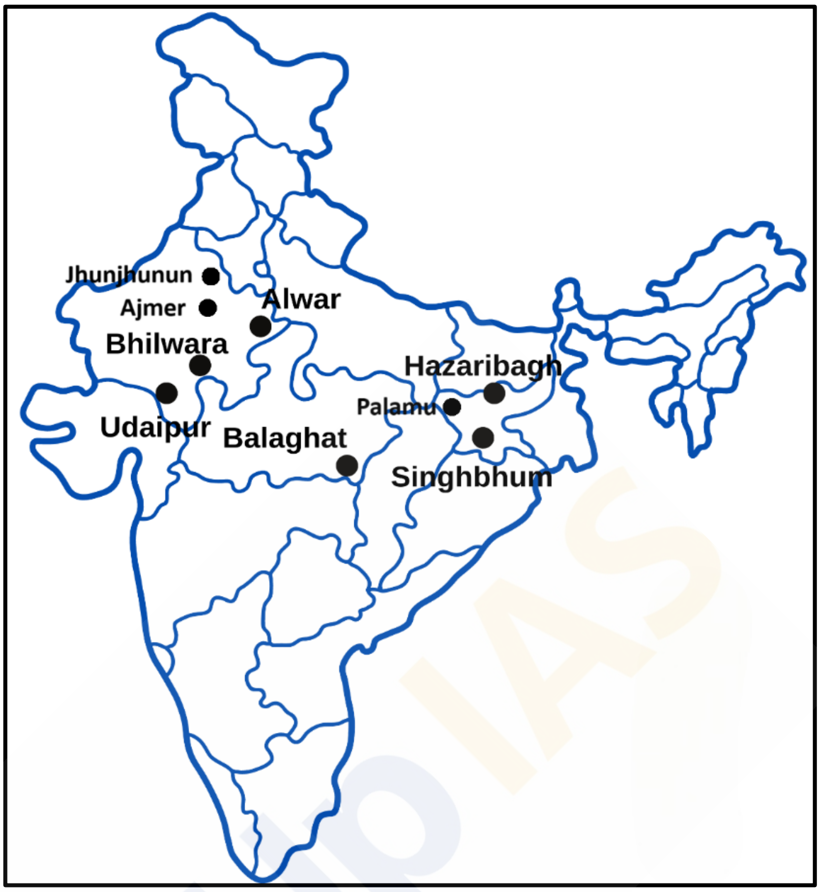
Copper:

- Copper has been one of the **earliest metals to come into human use,** predating iron by centuries. Its utility spans back to ancient times when it was commonly used for making **utensils** and **coins**. Its properties as a metal, such as being a **good conductor of electricity** and **ductile**, have made it indispensable in a wide variety of applications, particularly in the **electrical industry**. Today, copper is extensively utilized in **electrical machinery**, **wires**, and **cables**. Moreover, copper plays a significant role in the **automobile** and **defence industries**, owing to its versatility and durability.
- Beyond its standalone use, copper is often alloyed with other metals to produce materials with enhanced properties. For instance, it is combined with **iron** and **nickel** to create **stainless steel**, with **nickel** to form **Monel metal**, and with **aluminium** to produce **duralumin**. Additionally, copper forms **brass** when alloyed with **zinc** and **bronze** when mixed with **tin**. These alloys are crucial in various industrial applications due to their strength, corrosion resistance, and conductivity.
- Copper ore is found in both **ancient** and **younger rock formations**, typically occurring as **veins**, **disseminations**, and **bedded deposits**. However, **mining copper** is a **costly** and **labour-intensive** process, primarily because most copper ores contain a **low percentage** of the metal. Globally, the average metal content in copper ore is around **2.5%**, but in India, the **ore grade** is less than **1%**, making extraction more challenging and expensive.
- **India** has significant copper reserves, but it is **not self-sufficient in copper production** and imports a substantial amount of copper from countries like **Zimbabwe**, **Australia**, **USA**, **Mexico**, and **Japan**.

Production and Distribution of Copper in India:

- India's copper reserves are concentrated mainly in three states: **Rajasthan**, **Madhya Pradesh**, and **Jharkhand**. The distribution of copper ore reserves and production across these states varies.

Rajasthan:



- Rajasthan holds the **largest copper ore reserves** in India, with **50% of the country's total reserves**, though it contributes to only **28% of copper production.** The most important copper-producing area in Rajasthan is the **Khetri-Singhana belt** in the **Jhunjhunu District**. Other notable copper mining areas include **Ajmer**, **Alwar**, **Bhilwara**, **Chittorgarh**, **Dungarpur**, **Jaipur**, **Pali**, **Sikar**, and **Sirohi** districts. Additionally, the **Koh-Dariba Mountain** (located about 48 km from **Alwar city**) and the **Delwara-Kirovli area** (30 km from **Udaipur**) are other key copper-producing regions.

Madhya Pradesh:

- Madhya Pradesh is the **largest producer of copper** in India, contributing to **59% of the country's copper production**. The state holds **24% of India's total copper reserves**. Significant copper deposits are located in the **Taregaon** region of the **Malanjkhand belt** in the **Balaghat District**. Copper is also found in the **Bargaon** area of the **Betul District**.

Jharkhand:

- Jharkhand is the **third largest copper-producing state** in India, with **19% of the country's copper reserves** and contributing **11% to the nation's copper production**. Copper is mined in **Hazaribagh** and **Santhal Pargana** districts of Jharkhand. In the neighboring state of Bihar, copper is also extracted from **Gaya** and **Palamu** districts.

Copper Imports and Supply:

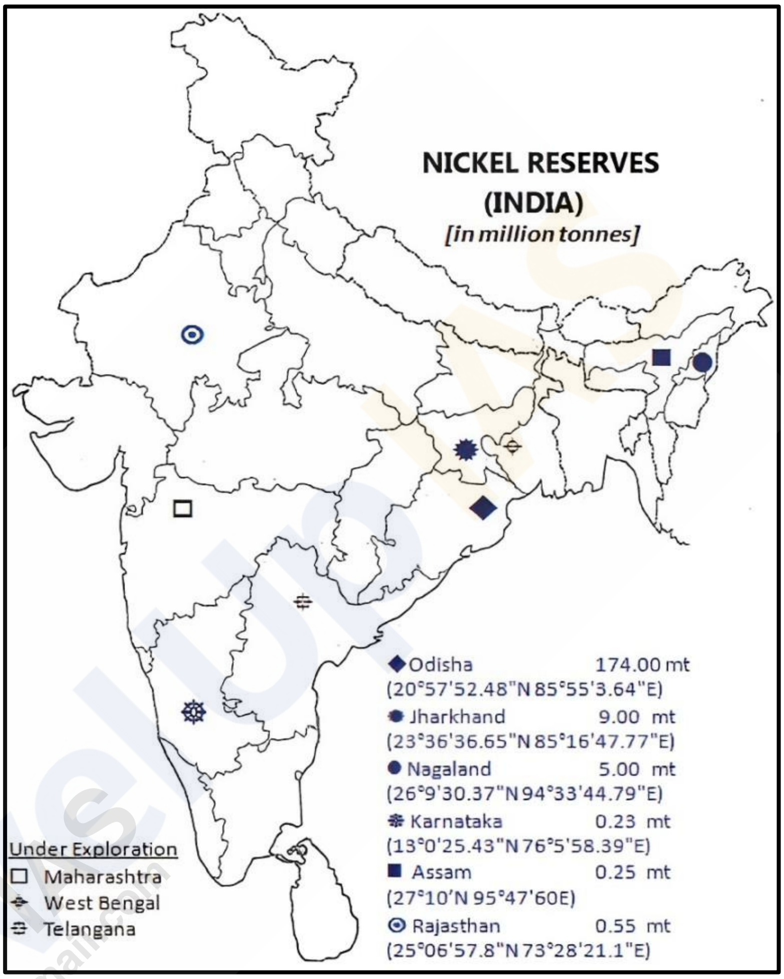
- India's domestic production of copper ore has historically been insufficient to meet its growing demand. As a result, the country is reliant on **copper imports** from other nations to bridge the gap. Key suppliers of copper to India include the **USA**, **Canada**, **Zimbabwe**, **Japan**, and **Mexico**. The quantity of copper imported varies each year, depending on the **demand** and **supply** dynamics within the global market.

Nickel:

- Nickel is a metal that **does not occur freely in nature** and is usually **found in association with other metals** such as **copper, uranium**, and others. It plays a significant role in various industries due to its versatile applications, particularly in the production of alloys.
- Nickel is an **important alloying material**. When alloyed with iron, it produces **rust-proof stainless steel** of superior quality, which is commonly used for manufacturing utensils. Due to its **greater hardness and tensile strength**, nickel steel is also used for making **armoured plates, motor cars, bullet jackets, and naval constructions**. In addition, when alloyed with **copper or silver**, nickel is used to produce **coins**.

- **Nickel-aluminum alloys** find applications in **Aeroplan manufacturing** and **internal combustion engines** due to their high strength and durability. **Metallic nickel** is also widely used for manufacturing **storage batteries** and serves as a **catalyst** in the **hydrogenation process** for hardening fats and oils, which are then used in the production of soap, foodstuffs, and **vanaspati**.

Production and Distribution of Nickel in India:



- Significant **nickeliferous limonite deposits** are found in the **Sukinda Valley** of **Jajapur district, Odisha**, where nickel occurs in **oxide form**. Processes are currently being developed to utilize this nickel effectively. Nickel also occurs in **sulphide form** along with copper mineralization in the **East Singhbhum district, Jharkhand**. In addition, nickel is found in association with **uranium deposits** at **Jadugoda, Jharkhand**, and methods are being developed for its recovery. Other notable occurrences of nickel have been identified in **Karnataka, Kerala,** and **Rajasthan**.
- A potential future source of nickel lies in **polymetallic sea nodules**, which contain significant amounts of the metal.
- The total **nickel ore resources** in India are estimated at **189 million tonnes**, with **92%** of these resources, or **175 million tonnes**, located in **Odisha**. The remaining **8%** of resources are distributed across **Jharkhand** (9 million tonnes) and **Nagaland** (5 million tonnes), with nominal amounts reported from **Karnataka** (0.23 million tonnes).
- **Nickel's strategic importance,** especially in alloy production, defense, and industrial applications, makes it a valuable resource in India's mineral economy.

Lead and Zinc:

Lead:

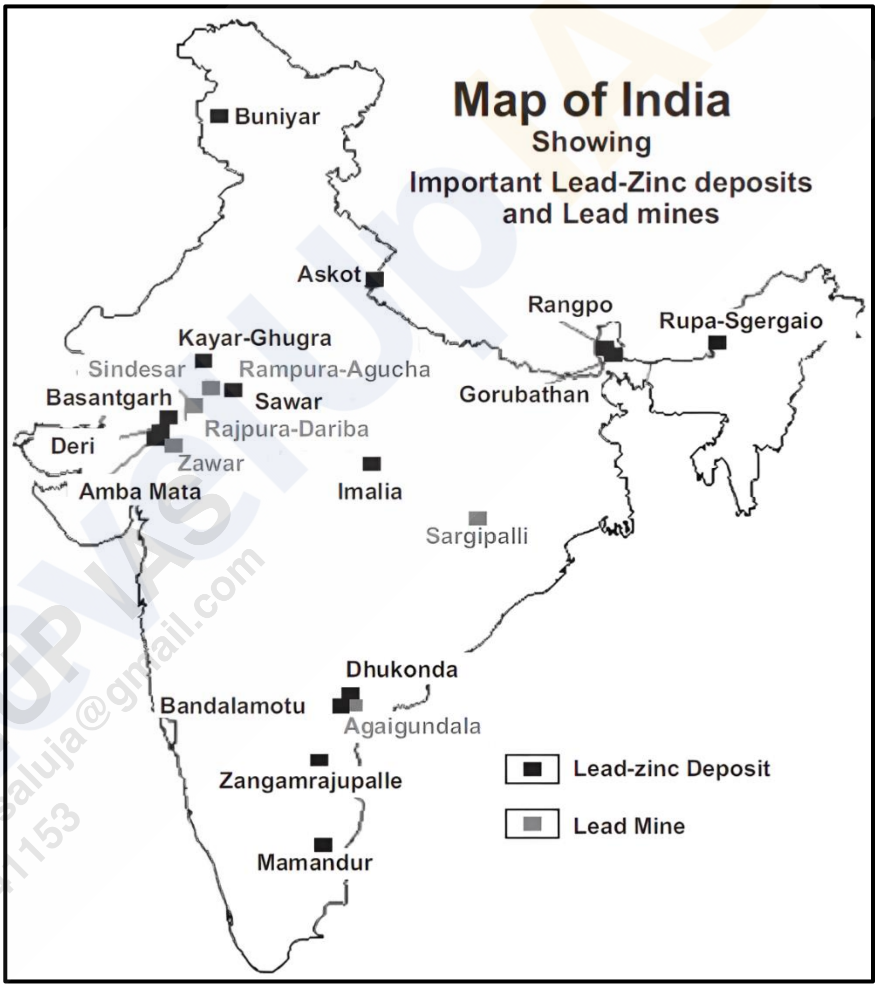
- Lead is a highly **versatile metal**, valued for its **malleability, softness, heaviness,** and **poor conductivity of heat**. Its most significant industrial application is as a **constituent in alloys** such as type metal, bronzes, and anti-friction metals. **Lead oxide** finds wide usage in industries, particularly in the production of **lead sheeting, cable covers, ammunition, paints, glassmaking**, and in the **rubber industry**. Moreover, **lead is formed into sheets, tubes, and pipes**, which are extensively used in construction, particularly for **sanitary fittings**. In recent times, lead has gained prominence in the **automotive and aerospace industries** as well as in **calculating machines**. **Lead nitrate** is another important compound, used extensively in the **dyeing and printing industries**.

- In nature, lead does not occur in a free form. Instead, it is typically found as **galena**, a cubic **sulphide** mineral (PbS). **Galena** occurs in **veins** within **limestones, calcareous slates, sandstones,** and occasionally in **metamorphic rocks**. It can also be found in **volcanic rock associations**.

Zinc:

- Zinc is often found in **mixed ores** containing both **lead and zinc**. The ore is found in veins, usually associated with **galena, chalcopyrites, iron pyrites,** and other **sulphide ores**. Zinc has broad industrial applications, with its primary use being for **alloying** and the manufacture of **galvanized sheets**. In addition, zinc is crucial for producing **dry batteries, white pigments, electrodes, textiles,** and **die-casting products**. Zinc is also widely used in the **rubber industry** and for manufacturing **collapsible tubes** that store drugs, pastes, and similar materials.

Production and Distribution of Lead and Zinc in India:



- **India** is endowed with substantial resources of lead and zinc. The **total resources of lead and zinc ores** are estimated at around **750 million tonnes.** Out of this, 118.98 million tonnes (16%) are classified under 'reserves', while the remaining 592.61 million tonnes are categorized as 'remaining resources'. In terms of metal content, lead resources are estimated at 20 million tonnes, while zinc accounts for 36.66 million tonnes. Additionally, there are 118.45 thousand tonnes of combined lead and zinc metal resources.

Rajasthan:

- Rajasthan is a major hub for lead and zinc mining. The metals are extracted from the **Zawar mines** in the **Udaipur district** and from **Angunche Rampura** in the **Bhilwara district**. These mines also produce **silver ore** as a byproduct.

Andhra Pradesh:

- In Andhra Pradesh, significant reserves of lead and zinc are found in the **Cuddapah district**.

Gujarat:

- In Gujarat, lead and zinc are mined from regions such as **Banaskantha**, **Vadodara**, **Panchmahal**, and **Surat**.

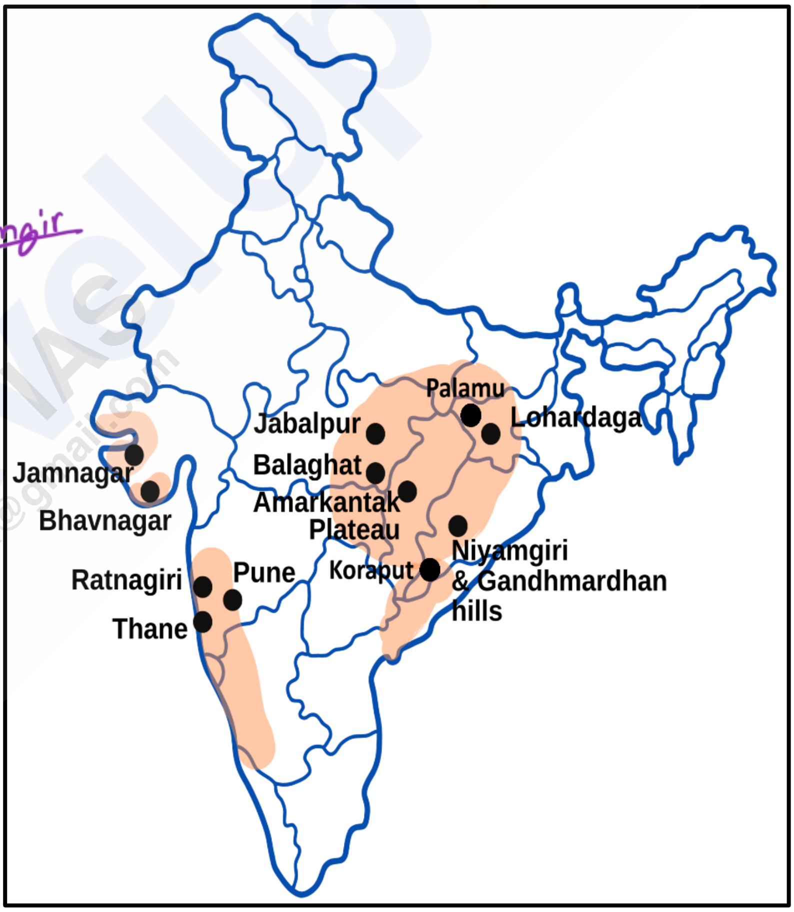
Bauxite:

- **Bauxite** is a vital ore used in the **production of aluminium,** which is one of the most commonly used metals globally. It is primarily composed of **hydrated aluminium oxides** and is not classified as a single mineral, but rather a rock. The majority of **bauxite deposits** are associated with **lateritic regions** and are typically found as capping on hills and plateaus, except in coastal regions like Gujarat and Goa.
- The aluminium industry can be divided into **two major segments - Alumina Production Plants** and **Aluminium Reduction Plants**. The former processes **bauxite ore** to extract **alumina**, while the latter **reduces alumina** into **aluminium.** Typically, alumina production plants are located near bauxite mines, and aluminium reduction plants are situated near sources of cheap electricity due to the high energy demands of the process. **To produce one ton of aluminium,** approximately **six tons of bauxite** is required, which yields around **two tons of alumina**.
- The extraction process begins with the treatment of **bauxite ore** using concentrated **sodium hydroxide**, resulting in the formation of **soluble sodium aluminate**. This is then filtered off, and upon heating with water, **aluminium hydroxide** is produced. Upon further strong heating, aluminium hydroxide decomposes to form **alumina**.

Production and Distribution of Bauxite in India:

Orissa:

- Orissa is the **largest bauxite-producing state** in India, contributing nearly **50% of the country's total production**. The main bauxite belt stretches through the **Kalahandi, Koraput,** Bolangir and **Baragarh districts**. This belt, which extends 300 km in length, 40 to 100 km in width, and up to 1300 meters in thickness, holds India's largest bauxite reserves. Major deposits include the **Panchpatmali** deposits in the **Koraput district** and the **Gandha Mardan** deposits in the **Baragarh district.** Panchpatmali is particularly notable as it is considered the largest bauxite deposit in the country.



Gujarat:

- **Gujarat** is the **second-largest producer** of bauxite, accounting for over **15% of India's total production**. The most important deposits are located in a narrow belt between the **Gulf of Kachchh and the Arabian Sea**, extending through **Bhavnagar, Junagadh, and Amreli districts**.

Jharkhand:

- The bauxite reserves in **Jharkhand** are estimated at **63.5 million tonnes**. These reserves are primarily located in the **Ranchi, Lohardaga, Palamu,** and **Gumla districts**, with high-grade ore found in Lohardaga and adjacent areas. Small deposits also exist in **Dumka and Munger districts**.

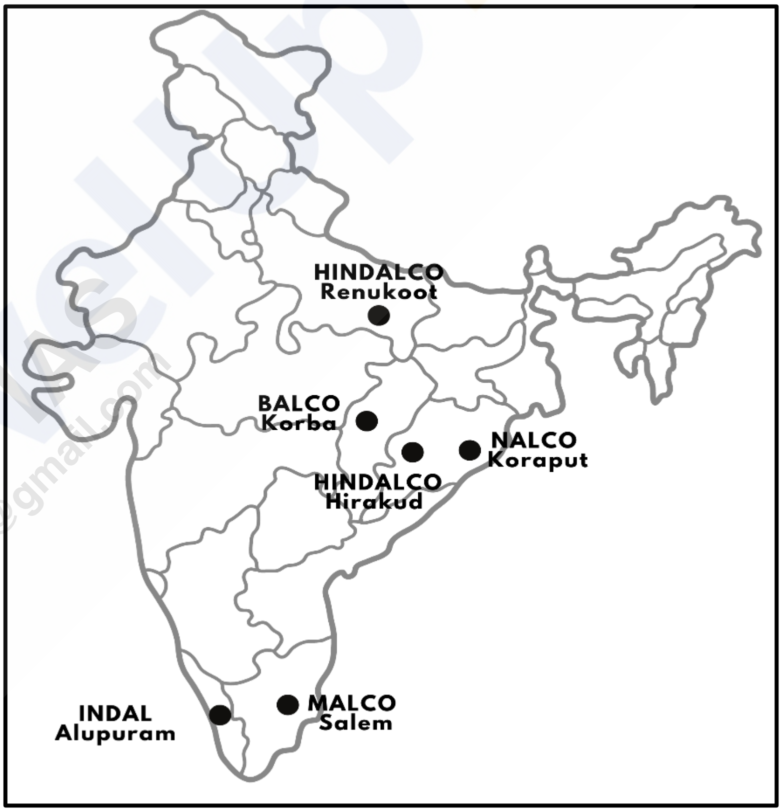
Other Bauxite-Producing States:

- **Maharashtra -** The **Kolhapur district** holds significant bauxite deposits, particularly on the plateau basalts.
- **Chhattisgarh -** Rich bauxite deposits are found in the **Maikala range** and on the **Amarkantak plateau** in **Bilaspur, Durg, Surguja, Raigarh, and Bilaspur**.
- **Tamil Nadu - Nilgiri and Salem** districts contribute over 2% of the country's total production.
- **Madhya Pradesh -** The **Amarkantak plateau**, **Maikala range**, and **Kotni area** of Jabalpur are important bauxite sources.
- Smaller bauxite deposits also occur in **Andhra Pradesh** (Vishakhapatnam, East and West Godavari), **Kerala** (Kannur, Kollam, Thiruvananthapuram), **Rajasthan** (Kota), **Uttar Pradesh** (Banda, Lalitpur, Varanasi), **Jammu and Kashmir** (Jammu, Poonch, Udhampur), and **Goa**.

Export of Bauxite from India:

- Approximately **80%** of India's bauxite is **used domestically** for aluminium production. However, **exports** have decreased due to rising domestic demand, though small amounts are still exported. **Italy** is the **largest importer** of Indian bauxite, receiving **60%** of exports, followed by the **UK** (25%), **Germany** (9%), and **Japan** (4%).

Aluminium Plants in India:



Hindalco Aluminium Plants:

- **Renukoot (Uttar Pradesh) -** This is Hindalco's largest aluminium plant, playing a key role in India's aluminium industry.
- **Hirakud (Orissa) -** Located on the banks of the Hirakud Dam, this plant was established by Indal in 1959 and is the second aluminium smelter in India.
- **Alupuram (Kerala) -** Although the smelter has been shut down, the extrusion unit at this plant is still operational. It is also the site where aluminium ingot was produced for the first time in India.

National Aluminium Company Ltd. (NALCO):

- **Angul (Orissa):** Nalco's Angul plant is a key player in the Indian aluminium industry, producing a significant amount of the country's aluminium.

Bharat Aluminium Company Ltd. (BALCO):

- **Korba (Chhattisgarh):** Set to become the **largest aluminium plant in the world**, this plant has a capacity of **1 million tonnes per annum**.

Madras Aluminium Company Ltd. (MALCO):

- **Mettur (Tamil Nadu):** Malco's Mettur plant is located at the Mettur Dam complex and contributes to the country's aluminium production.

Tungsten:

- Tungsten, a **highly valuable metal,** is primarily derived from its chief ore, **'wolfram.'** This metal exhibits several unique properties, making it indispensable across various industries. Among its most significant characteristics is its **self-hardening ability**, which makes it crucial for the **steel industry.** In fact, **over 95% of wolfram** is consumed by the steel sector. When tungsten is added in specific proportions to steel, it enhances its strength and durability, which is vital in the production of **ammunitions, armour plates, heavy artillery, and hard cutting tools**.
- Tungsten is **easily alloyed** with other metals like **chromium, nickel, molybdenum, and titanium**, resulting in **hard-facing, heat-resistant, and corrosion-resistant alloys**. These properties make tungsten alloys valuable for industrial applications beyond steel, such as in **electric bulb filaments, paints, ceramics, and textiles**.

Production and Distribution of Tungsten in India:

- India has significant tungsten resources, with total reserves estimated at **87.4 million tonnes** of ore containing **142,094 tonnes of tungsten (WO_3)**. These resources are unevenly distributed across various states:
  - **Karnataka** accounts for **42%** of the total resources.
  - **Rajasthan** holds **27%** of the reserves.
  - **Andhra Pradesh** possesses **17%** of the total resources.
  - **Maharashtra** accounts for **9%** of the reserves.
  - The remaining **5%** of the resources are spread across **Haryana, Tamil Nadu, Uttarakhand, and West Bengal**.
- In **Degana, Rajasthan**, the **WO_3 content** in vein deposits ranges from **0.25% to 0.54%**, while in gravel deposits, the concentration is around **0.04%**. In **Bankura, West Bengal**, the deposit contains an average of **0.1% WO_3**. Furthermore, the **Kuhi-Khobana-Agargaon belt** in the **Sakoli Basin** (Maharashtra) has shown promising tungsten potential. The Geological Survey of India (GSI) has identified **seven mineralized zones** in **Bhandara** and **Nagpur** districts. In the **Kuhi block**, the WO_3 content ranges from **0.01% to 0.19%**, while in the **Khobana block**, the range is **0.13% to 0.38%**. The **Pardi-Dahegaon-Pipalgaon block** contains an average of **0.48% WO_3**, with the overall deposit averaging **0.17% WO_3**.
- In **Karnataka, gold ore** from the **Mysore mine** of **Bharat Gold Mines Limited (BGML)** has been identified as a potential source of **scheelite** (a tungsten mineral). Additionally, tailing dumps at the **Kolar Gold Fields** contain WO_3 values ranging from **0.035% to 0.18%**.
- **India's yearly tungsten ore production** is estimated at **45,000 to 50,000 kg**. However, this output is insufficient to meet domestic demand, leading the **country to rely on imports** to bridge the gap in tungsten requirements. Imports are **mainly from Singapore, USA & UK.**

Pyrite:

- **Pyrite**, commonly known as **"fool's gold"** due to its **metallic luster and pale brass-yellow hue,** is a sulphide mineral of **iron (FeS_2)**. Despite containing iron, it is **not valued as an ore of iron because its high sulphur content** makes it **unsuitable for iron extraction.** The primary economic value of pyrite lies in being a **chief source of sulphur**.
- Sulphur extracted from pyrite is essential in producing **sulphuric acid**, used in industries like **fertilizers, chemicals, petroleum, steel**, and **rayon**. Additionally, elemental sulphur is crucial for manufacturing **explosives, matches, insecticides, fungicides**, and in **vulcanizing rubber**.

Production and Distribution of Pyrite in India:

- Pyrite is widely distributed and is found in geological formations ranging from the oldest **crystalline rocks** to the **youngest sediments.** Pyrites occur in **Son Valley** (Bihar), **Chitradurga** and **Uttar Kannada** (Karnataka), **Assam** coalfields, **Sikar** (Rajasthan), Garhwal, Tehri Garhwal and Almora (Uttarakhand) etc. Despite its wide distribution, the production of pyrites in India has experienced a significant decline over the years.

Gold:

- Gold is a **valuable metal** that holds significant importance **both economically and culturally.** It typically occurs in **auriferous rocks**, which are rocks or minerals containing gold. Additionally, it is found in **river sands,** known as **alluvial gold.** This precious metal has been widely used for **making ornaments** and serves as an **international currency** due to its universal acceptance and value.
- India's domestic **gold production is insufficient**, leading to a **reliance on imports** from countries like **Australia, Canada, and Myanmar.** Globally, significant gold deposits are found in countries such as **South Africa, Australia, Indonesia, Canada, Ghana, Chile, China, USA, and Russia**. These nations play a major role in meeting the global demand for gold.
- India possesses notable gold reserves, categorized based on **metal ore (primary)** and **metal content**. The primary gold ore resources are located in: **Bihar** (45%), **Rajasthan** (23%), **Karnataka** (22%), **West Bengal** (3%), **Andhra Pradesh** and **Madhya Pradesh** (2% each).
- In terms of **metal content**, significant reserves are found in **Karnataka, Rajasthan, Bihar, Andhra Pradesh, and Jharkhand**. The most prominent gold fields in India include the **Kolar Gold Field, Hutti Gold Field, and Ramgiri Gold Field.**

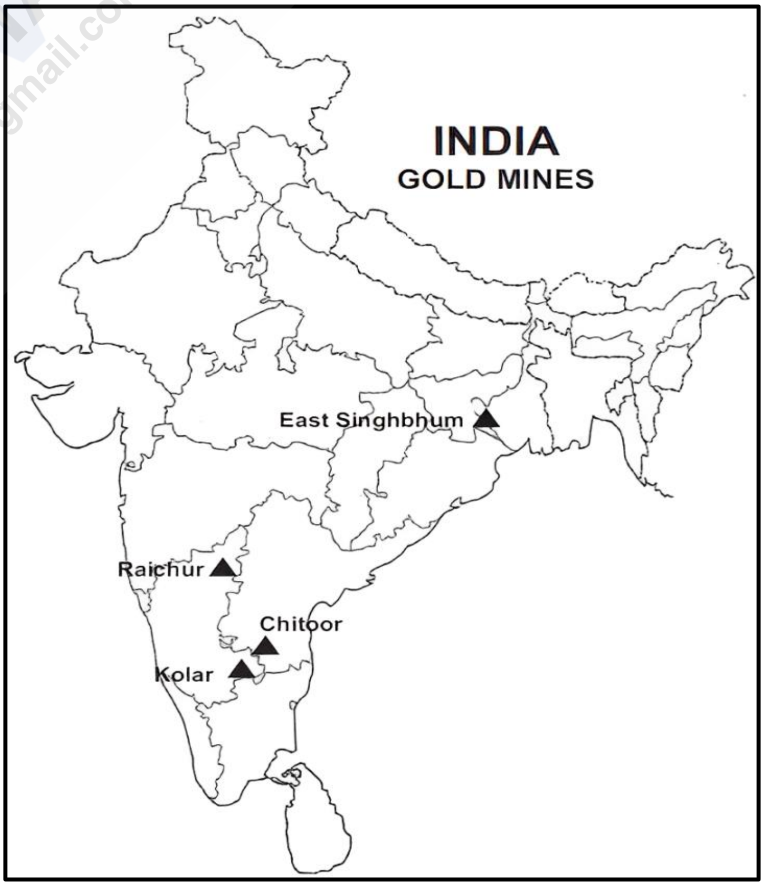
Production and Distribution of Gold in India:

Karnataka:

- Karnataka is the **largest producer** of gold in India, contributing approximately **80% of the country's total gold production**. Every district in Karnataka has some gold reserves, but the most important is the **Kolar Gold Field**, which has long been the largest gold supplier in India. Following Kolar, the **Hutti Mine** in the **Raichur** district is significant, although its ore is of **low grade** and its production is much lower than that of Kolar.

Andhra Pradesh:

- Andhra Pradesh is the **second-largest producer** of gold in India, though it significantly lags behind Karnataka. The primary deposits are found in **Ramagiri** in the **Anantapur district**, but this field is almost exhausted. Gold is also obtained from **placer deposits** in the sands of rivers in Andhra Pradesh.



Jharkhand:

- Jharkhand accounts for **about 10% of India's total gold production**, making it the **third-largest producer.** Gold in Jharkhand is found both in **placer deposits** along rivers like **Subarnarekha** and **Sonanadi**, and as **native gold** in the **Singhbhum District** and parts of the **Chhota Nagpur plateau**.

Other Gold Producing States:

- In **Kerala**, small amounts of alluvial gold are found in the **river terraces** along the **Punna Puzha** and **Chabiyar Puzha**.

Major Gold Mines in India:

- India is home to several significant gold mines. The **Hatti Gold Mines** and **Kolar Gold Fields**, both located in Karnataka, are among the most prominent gold-producing sites. In **Jharkhand**, important mines include the **Lava Gold Mines**, **Parasi**, **Pahadia**, **Kunderkocha**, and **Bhitar Dari**. The **Sonbhadra Mine** in **Uttar Pradesh** also contributes to India's gold production. These mines, though scattered across different states, play a vital role in the country's gold mining industry.

Silver:

- **Silver** is a highly valued precious metal in India, extensively used in **religious ceremonies, festivals, weddings,** and other cultural events. Its importance comes **second only to gold in terms of ornament-making and traditional use.** Silver is typically found mixed with other metals such as **copper, lead, gold, zinc,** and others.
- Silver has a wide range of industrial applications. It is used in the manufacture of **chemicals**, in **electroplating**, **photography**, and for **colouring glass**. Despite its importance, **India is not a major producer of silver** and relies heavily on imports to meet its demand.

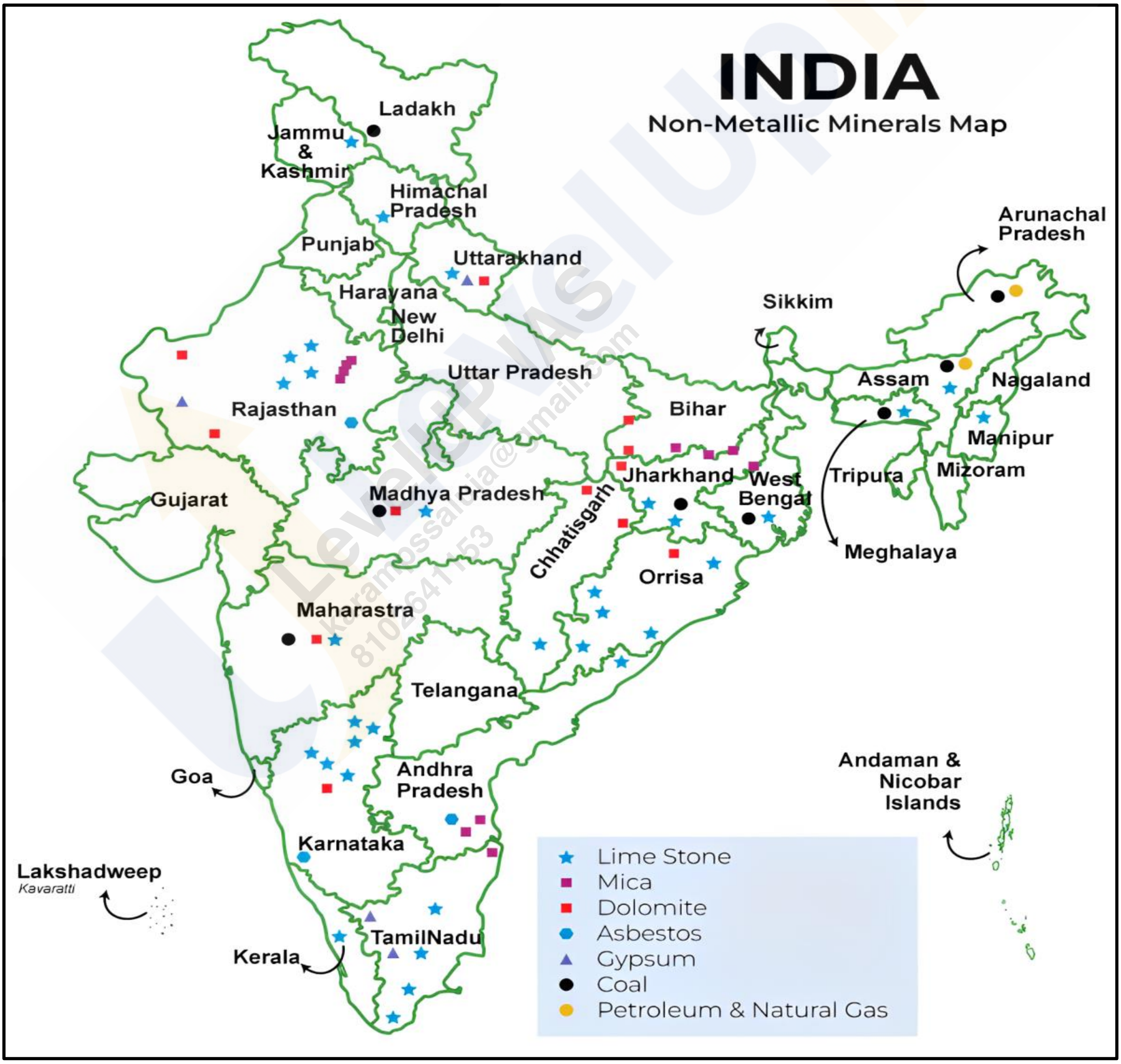
Production and Distribution of Silver in India:

- Silver is usually found in association with ores of **lead, zinc, copper,** and **gold**. It is often extracted as a **byproduct** during the electrolysis of these metals or through **chemical methods**. The largest silver production in India comes from **Rajasthan**, followed by **Gujarat** and **Jharkhand**.
- One of the key sources of silver is the **Zawar mines in Udaipur (Rajasthan)**, where it is obtained as a byproduct of **Galena ore** (lead) at the **Hindustan Zinc Smelter**. Other sources include the **Kolar Goldfields** and **Hutti Gold Fields**, where silver is produced during the gold refining process. Additionally, the **Tundoo lead smelter** in the **Dhanbad district (Jharkhand)** produces silver as a byproduct of lead refining. The **Vizag Zinc smelter** in **Andhra Pradesh** also produces silver from lead concentrates.

- Silver traces have been found in states such as **Jharkhand, Andhra Pradesh, Gujarat, Karnataka, Jammu and Kashmir,** and **Uttarakhand**.
- The major silver mines in India include the **Sindesar Khurd Mine** in **Rajasthan**, the **Rampura Agucha Mine** in **Rajasthan**, the **Zawar Mine** in **Rajasthan**, the **Rajpura Dariba Mine** in **Rajasthan**, and the **Hutti Mine** in **Karnataka**.

Non-Metallic Minerals:

- **India** is rich in a diverse array of **non-metallic minerals**, which, while not as economically significant as metallic minerals, play a **crucial role in various industries.** These minerals are fundamental to numerous sectors, contributing to the country's industrial landscape.
- Although only a limited number of non-metallic minerals have gained substantial industrial and economic importance, their applications span a wide range of industries. The most notable sectors utilizing these minerals include **cement production**, **fertilizer manufacturing**, and the **electrical industry**. In the cement industry, non-metallic minerals serve as key components in the production process, while fertilizers rely on various non-metallic minerals to enhance agricultural productivity. Additionally, the electrical sector uses non-metallic minerals for insulation and other essential functions.



Mica:

- Mica has been utilized in India since **ancient times**, primarily in **Ayurveda**, where it is referred to as **abhrak**. With the advent of the **electrical industry**, mica found new avenues for application, particularly due to its excellent **insulating properties**. This mineral is highly valued in the **electrical and electronics industry** because it can **withstand high voltage** and exhibits a **low power loss factor**.
- Mica is classified as a naturally occurring **non-metallic mineral**, composed of a collection of **silicates**. It typically occurs in the **veins of metamorphic rocks** and is also found in **igneous** and **sedimentary rocks**. Notably, mica is chemically **inert**, stable, and does not absorb water. Its unique combination of **elasticity, toughness, flexibility,** and **transparency** enhances its utility across various applications. Apart from its industrial uses, mica is incorporated in **toothpaste** and **cosmetics** due to its **glittery appearance** and also serves as a **mild abrasive** in toothpaste formulations.

Production and Distribution of Mica in India:

Andhra Pradesh:

- Contributes **41%** of the country's total mica resources, with the **Nellore district** noted for producing the best quality mica. Other prominent districts include **Vishakhapatnam, West Godavari,** and **Krishna**.

Rajasthan:

- The second largest producer, with a mica belt extending approximately **320 kilometers** from **Jaipur to Bhilwara**, including **Udaipur**. The **Bhilwara district** is the most important producer within this belt.

Jharkhand:

- Ranked as the third largest producer, mica is found in a belt that extends about **150 kilometers** in length and **32 kilometers** in width, stretching from the **Gaya district** of Bihar to **Hazaribagh** and **Koderma** districts in Jharkhand. This belt, which runs east-west, contains the richest deposits of high-quality **ruby mica**. Key production centers include **Kodarma** (home to the largest mica reserves in the world), **Giridih**, and **Domchanch**. Additional mica deposits are located in **Munger, Bihar**.

Other Mica Producing States:

- **Karnataka** (Mysuru and Hasan), **Tamil Nadu** (Coimbatore, Tiruchirapalli, Madurai, and Kanyakumari), **Kerala** (Alleppey), **Maharashtra** (Ratnagiri), **West Bengal** (Purulia and Bankura).

Exports of Mica from India:

- **India** stands as the **largest exporter** of mica, with certain grades being vital to the global electrical industries. The majority of mica exports are facilitated through the ports of **Kolkata** and **Vishakhapatnam**. Key importing countries include **Japan** (19%), the **USA** (17%), and the **U.K.**, among others. The significance of mica in various industries highlights India's crucial role in the **global mica market**.

Limestone:

- Limestone is primarily composed of **calcium carbonate** and **magnesium carbonate**. In addition to these main constituents, limestone also contains small amounts of **silica**, **alumina**, **iron oxides**, **phosphorus**, and **sulphur**. The two most significant components are **calcite** and **dolomite**. Limestone deposits are of **sedimentary origin** and can be found in all geological sequences from the **Pre-Cambrian** era to the **Recent**, with the exception of the **Gondwana** period.
- Limestone serves a variety of purposes across different industries. Approximately **75%** of total limestone consumption is utilized in the **cement industry**, while **16%** is used in the **iron and steel industry**. Additionally, **4%** is consumed by the **chemical industries**. The remaining limestone is employed in various other sectors, including **paper**, **sugar**, **fertilizers**, **glass**, **rubber**, and the **ferromanganese** industries.

Production and Distribution of Limestone in India:

- Limestone holds a preeminent position among non-fuel solid mineral deposits in India, particularly in terms of the volume of annual extraction. With **rapid urbanization** and an **increasing demand for housing and infrastructure,** the need for limestone is expected to rise further.
- The distribution of limestone resources in India is varied across different states. **Karnataka** leads with a significant share, holding **28%** of the total limestone resources. Following Karnataka, **Andhra Pradesh**, **Gujarat**, and **Rajasthan** each account for **11%** of the resources. **Telangana** contributes **9%**, while **Chhattisgarh** holds **5%**. **Madhya Pradesh** accounts for **4%** of the resources. The remaining **21%** of limestone resources is spread across various other states, showcasing the extensive geographic distribution of this essential mineral throughout the country.

Rajasthan:

- According to data from **2014-15**, Rajasthan was the top limestone-producing state, contributing **21% of the total production.** Key limestone-producing districts include **Jhunjhunu**, **Banswara**, **Jodhpur**, **Ajmer**, **Bikaner**, **Dungarpur**, **Kota**, **Tonk**, **Alwar**, **Sawai Madhopur**, and **Nagaur**.

Madhya Pradesh:

- Madhya Pradesh contributes around **13% of India's limestone production,** with significant production coming from districts such as **Jabalpur**, **Satna**, **Betul**, **Sagar**, **Damoh**, and **Rewa**.

Andhra Pradesh:

- Andhra Pradesh is responsible for approximately **12% of limestone production,** with major deposits located in **Cuddapah**, **Kumool**, **Guntur**, and **Krishna**.

Gujarat:

- Gujarat accounts for about **9% of the nation's limestone output.** The primary limestone-producing districts include **Amreli**, **Kachchh**, **Surat**, **Junagadh**, **Kheda**, and **Panchmahals**.

Chhattisgarh:

- Chhattisgarh produces around **8% of the total limestone in India,** with key deposits in the **Bastar**, **Bilaspur**, **Raigarh**, **Raipur**, and **Durg** districts.

Tamil Nadu:

- Also contributing about **8%** to limestone production, Tamil Nadu has major limestone-producing districts such as **Ramnathapuram**, **Tirunelveli**, **Tiruchchirappalli**, **Salem**, **Coimbatore**, **Madurai**, and **Thanjavur**.

Karnataka:

- Karnataka accounts for around **8% of limestone production,** with major districts including **Gulbarga**, **Chitradurga**, **Tumkur**, **Belgaum**, **Bijapur**, **Mysore**, and **Shimoga**.

Telangana:

- Telangana contributes approximately **8% to the limestone production** in India, with significant deposits found in **Nalgonda**, **Adilabad**, **Warangal**, **Mahabubnagar**, and **Karimnagar**.

Other States:

- The remaining **5%** of limestone production is attributed to states like **Meghalaya**, **Odisha**, **Uttar Pradesh**, **Jharkhand**, **Assam**, **Kerala**, **Bihar**, and **Jammu & Kashmir**.

Dolomite:

- Dolomite is an **anhydrous carbonate mineral** characterized by its composition of **calcium magnesium carbonate**, ideally represented by the chemical formula **$CaMg(CO_3)_2$**. This mineral also refers to a type of **sedimentary carbonate rock** primarily composed of dolomite. An alternative name used for this rock type is **dolostone**. When dolomite rock contains additional components, such as calcite or a mixture of calcite and magnesite, it is classified as **dolomitic limestone**.
- Dolomite serves several important industrial functions. It is mainly utilized as a **blast furnace flux** and as a source of **magnesium salts**. Furthermore, dolomite plays a vital role in the **fertilizer** and **glass industries**. Notably, the **iron and steel industry** is the largest consumer of dolomite, accounting for **90% of its usage,** followed by sectors such as **fertilizers**, **ferroalloys**, and **glass**.

Production and Distribution of Dolomite in India:

- Dolomite is **widely distributed** across India, with various states contributing to its production. In the fiscal year 2014-15, **Chhattisgarh** emerged as the **leading producer,** contributing **39% of the total production.** Other significant states include **Andhra Pradesh** (11%), **Karnataka** (10%), **Madhya Pradesh** (9%), **Telangana** (8%), **Odisha** (7%), **Gujarat** and **Rajasthan** (6% each). The remaining **4%** of production was shared by **Jharkhand**, **Maharashtra**, and **Uttarakhand**.

State-wise Dolomite Deposits:

- **Chhattisgarh -** The primary deposits are found in the **Bastar**, **Bilaspur**, **Durg**, and **Raigarh** districts.
- **Odisha -** Significant deposits occur in **Sundargarh**, **Sambalpur**, and **Koraput** districts.
- **Jharkhand -** Dolomite is found in bands located to the north of **Chaibasa**, particularly in **Singhbhum** and **Palamu** districts.
- **Rajasthan -** Major producing districts include **Ajmer**, **Alwar**, **Bhilwara**, **Jaipur**, and **Jaisalmer**.
- **Karnataka -** Key deposits are located in **Belgaum**, **Bijapur**, **Chitradurga**, and **Mysore** districts.

Asbestos:

- **Asbestos** refers to a group of **six naturally occurring fibrous silicate minerals** that have significant commercial value due to their **unique physical and chemical properties.** It is recognized for its **fibrous character, infusibility, low heat conductivity, and high resistance to electricity, sound, and corrosion by acids**. These properties make asbestos highly desirable in various industrial applications.
- **Two distinct mineral varieties** are classified under asbestos. One is a variety of **Amphibole**, and the other, more commercially important, is a fibrous variety of **serpentine known as Chrysotile**. **Chrysotile**, which constitutes around **80% of the asbestos used commercially**, is valued for its **filaments with high tensile strength** and its remarkable resistance to fire.
- Asbestos is used in a wide range of industries due to its **resistance to heat and fire**. Some of its major applications include the production of **fire-proof cloth, rope, paper, millboard, and sheeting**. It is also used in manufacturing **protective gear** such as aprons and gloves, and **brake linings** in automobiles.
- In the construction sector, **asbestos cement products**, including **sheets, pipes, and tiles**, are popular for building purposes due to their durability and fire-resistant properties. When asbestos is brittle, it is employed to create **filter pads** used for filtering acids. Additionally, asbestos is mixed with magnesia to form **magnesia bricks**, which are commonly used for **heat insulation**.

Production and Distribution of Asbestos in India:

- India has significant asbestos resources, with **Rajasthan** and **Karnataka** accounting for the majority of the reserves. **Rajasthan** holds approximately **13.6 million tonnes**, which is **61 percent** of the country's total resources, and **Karnataka** follows with **8.28 million tonnes**, or **37 percent**. The remaining **2 percent** of the resources are distributed among **Jharkhand, Andhra Pradesh, Odisha, and Uttarakhand**.
- Key asbestos locations in India are distributed across several states. In **Rajasthan**, asbestos is found in the districts of **Udaipur, Dungarpur, Alwar, Ajmer, and Pali**. **Karnataka** holds significant deposits in the districts of **Hassan, Mandya, Shimoga, Mysore, and Chikmaglur**. In **Andhra Pradesh**, fine-quality asbestos is primarily located in **Pulivendla Taluk** of **Cuddapah district**. **Jharkhand** has asbestos resources concentrated in the **Singhbhum district**, while **Uttarakhand** has occurrences in the **Chamoli district**. These regions play an important role in asbestos production within India.

Magnesite:

- **Magnesite** is a **carbonate mineral** composed primarily of **magnesium carbonate ($MgCO_3$).** It is commonly formed as an alteration product of **dunites (peridotite)** and other basic magnesian rocks through geological processes. This mineral has a wide range of industrial applications, making it a valuable resource in several sectors.
- Magnesite is primarily utilized in the **manufacturing of refractory bricks**, which are **essential for high-temperature industrial processes,** particularly in **furnaces and kilns.** It also serves as a **bonding material in abrasives** and plays a role in the **production of special types of cement** used for artificial stone and tiles. Additionally, magnesite is an important raw material for the **extraction of magnesium metal**. The **steel industry** also heavily relies on magnesite for its operations, given its high-temperature resistance and other favourable properties.

Production and Distribution of Magnesite in India:

- India is home to significant magnesite deposits, with **Tamil Nadu** being the **largest producer.** One of the world's most notable magnesite deposits, and India's largest, is located at the **Chalk Hills near Salem town in Tamil Nadu**. In addition to Tamil Nadu, other states with magnesite resources include **Uttarakhand, Rajasthan, Andhra Pradesh, Himachal Pradesh, Jammu & Kashmir, Karnataka**, and **Kerala**. Tamil Nadu, Uttarakhand, and Karnataka are the leading producers of magnesite in the country.

Kyanite:

- Kyanite is a significant mineral that primarily occurs in **metamorphic aluminous rocks**. Its unique ability to withstand high temperatures makes it valuable in various industries, such as **metallurgical, ceramic, refractory, electrical, glass, and cement** industries. Additionally, kyanite is used in the production of **spark plugs for automobiles**.
- **India holds the largest kyanite deposits globally**, with all three grades of the mineral available in the country. These grades are determined by their **aluminium content**, with higher aluminium levels signifying higher quality. The states of **Jharkhand, Maharashtra, and Karnataka** produce almost the entirety of India's kyanite.
- Key deposits are located in **Saraikela (Jharkhand), Bhandara and Nagpur districts (Maharashtra)**, and in **Chikmaglur, Chitradurga, Mandya, Mysore, Dakshin Kannad, and Shimoga districts** (Karnataka). In terms of state-wise resource distribution, **Telangana** contributes **47%**, followed by **Andhra Pradesh (31%)**, **Karnataka (13%)**, and **Jharkhand (6%)**. The remaining **3% of resources** are distributed across **Kerala, Maharashtra, Rajasthan, Tamil Nadu, and West Bengal**.

Sillimanite

- The mineral **sillimanite** shares similar occurrences and uses with kyanite. It is widely utilized in the same industries, benefiting from its resistance to high temperatures. Major concentrations of sillimanite are found in **Tamil Nadu, Odisha, Kerala, Andhra Pradesh, and West Bengal**.
- Significant deposits of sillimanite in India are located in **Ganjam district (Odisha)**, **Kozhikode, Palakkad, Ernakulam, and Kottayam districts (Kerala)**, **Bhandara district (Maharashtra)**, **Udaipur district (Rajasthan)**, and **Hassan, Mysore, and Dakshin Kannada districts (Karnataka)**. The state-wise distribution shows **Tamil Nadu** with **26%** of the total resources, followed by **Odisha (20%)**, **Uttar Pradesh (17%)**, **Andhra Pradesh (14%)**, **Kerala (11%)**, and **Assam (7%)**. The remaining **5% of resources** are spread across **Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Meghalaya, Rajasthan, and West Bengal**.

Gypsum:

- **Gypsum** is a hydrated sulfate of calcium and appears as a **white opaque or transparent mineral**. It forms in **sedimentary formations** like limestones, sandstones, and shales. Gypsum has numerous applications, most notably in the manufacture of **ammonium sulfate fertilizers** and in the **cement industry**, where it constitutes **4-5% of cement**. Additionally, gypsum is used in making **plaster of Paris, ceramic molds, tiles, and plastics**. In agriculture, it is applied as surface plaster to conserve moisture and facilitate nitrogen absorption in soil.
- **Rajasthan** is the largest producer of gypsum in India, accounting for **81%** of total production. The primary deposits are found in the **Tertiary clays and shales of Jodhpur, Nagaur, and Bikaner**. Other regions with gypsum-bearing rocks include **Jaisalmer, Barmer, Chum, Pali, and Ganganagar**. Besides Rajasthan, gypsum is also produced in **Tamil Nadu (Tiruchirapalli district)**, **Jammu & Kashmir**, **Gujarat**, and **Uttar Pradesh**. State-wise, Rajasthan holds **81% of resources**, followed by **Jammu & Kashmir (14%)** and **Tamil Nadu (2%)**. The remaining **3%** of resources are located in **Gujarat, Himachal Pradesh, Karnataka, Uttarakhand, Andhra Pradesh, and Madhya Pradesh**.

Potash:

- **Potash** is the common name for **fertilizer-grade potassium (K)**, which is naturally abundant. However, in India, occurrences of potash minerals are not commercially exploitable, so **no domestic production** takes place. As a result, India meets its potash needs entirely through **imports**.
- India's potash resources are mainly found in **Rajasthan (94%)**, followed by **Madhya Pradesh (5%)** and **Uttar Pradesh (1%)**. Key deposits are located in **Satna & Sidhi districts (Madhya Pradesh)**, **Sonbhadra district (Uttar Pradesh)**, and **Jaisalmer, Chittorgarh, and Kota districts (Rajasthan)**.

Sulphur:

- India does not have mineable **elemental sulphur reserves**. Historically, **pyrites** were used as a sulfur substitute for producing sulfuric acid, but production ceased after 2003. Today, **elemental sulphur** is produced domestically as a **by-product of petroleum refineries** and **fuel oil** used in fertilizer production.

Salt:

- **Salt** in India is sourced from **seawater, brine springs, wells, salt pans**, and rocks. India ranks as the **third-largest salt producer in the world**, following the US and China. **Rock salt** is extracted in the **Mandi district (Himachal Pradesh)** and **Gujarat**, with the **Gujarat coast** contributing nearly half of the country's salt production. Other major salt-producing states include **Tamil Nadu, Rajasthan, Andhra Pradesh, Maharashtra, West Bengal, Karnataka, Odisha, and Goa**.

Diamond:

- **Diamond** has been prized as one of the **most valuable gems for over 2,000 years.** It is the **hardest naturally occurring substance found on Earth,** making it unique and highly sought after. Diamonds primarily occur in **two types of deposits:** Igneous rocks of basic or ultrabasic composition, and **Alluvial deposits** that are derived from primary sources.
- Diamonds are **formed deep within the Earth's mantle** and brought to the surface through **volcanic activity**. This transportation often results in diamonds being deposited in dykes, sills, and other geological formations. Diamonds are composed of **pure carbon** and **crystallize** in a **cubic system**, with the **octahedron** being the most common crystal form.

Diamond Industry in India:

- **India has a long history of diamond cutting and polishing,** especially for smaller diamonds. In fact, India is a global leader in this field, with **80% of the world's polished diamonds being processed in the country,** particularly in **Surat, Gujarat**. The Indian diamond industry is renowned for its expertise and scale, contributing significantly to the global market.
- Apart from their **ornamental value,** diamonds have extensive **industrial applications**. Being the hardest natural substance, diamonds are used in **grinding, drilling, cutting, and polishing tools**. Additionally, diamonds exhibit the highest **thermal conductivity** among minerals and possess high **electrical resistivity**, making them suitable for use in **semiconductors** and other technological applications.

Production and Distribution of Diamond in India:

- Diamond deposits in India have been known since prehistoric times. **The major diamond fields in India are grouped into four regions:**
    - **South Indian Tract (Andhra Pradesh) -** This region includes parts of **Anantapur, Kadapa, Guntur, Krishna, Mahabubnagar**, and **Kurnool** districts.
    - **Central Indian Tract (Madhya Pradesh) -** The **Panna belt** is the primary area of interest here.
    - **Chhattisgarh Tract -** This includes the **Behradin-Kodawali area** in Raipur district and regions like **Tokapal** and **Dugapal** in Bastar district.
    - **Eastern Indian Tract (Odisha) -** This region lies between the **Mahanadi and Godavari valleys**.
- Of these, diamond reserves have been estimated primarily in the **Panna belt** of **Madhya Pradesh** and **Krishna Gravels** of **Andhra Pradesh.** Recently, new **kimberlite fields** were discovered in the **Raichur-Gulbarga districts** of Karnataka.
- Madhya Pradesh holds the largest share of diamond resources in India, accounting for **90.18%** of the total. Andhra Pradesh follows with **5.72%**, and **Chhattisgarh** has **4.09%** of the country's diamond resources.
- India has only one active diamond mine, located at **Majhgaon in Panna** (Madhya Pradesh), which is operated by the **National Mineral Development Corporation (NMDC)**. The mine has a production capacity of **84,000 carats**, and to date, it has produced over **1 million carats** of diamonds.

Graphite:

- Graphite is a **naturally occurring form of crystalline carbon** and is also referred to as **plumbago**, **black lead**, or **mineral carbon**. It is the most stable form of naturally occurring carbon and contains **no less than 95% carbon**. Graphite is **often considered the highest grade of coal,** situated just **above anthracite** in terms of carbon content. However, it is not commonly used as a fuel because it is difficult to ignite. Graphite is primarily found in **metamorphic** and **igneous rocks**, and it has unique physical properties, being **extremely soft** and easily cleaving with minimal pressure. Additionally, it is **highly resistant to heat** and has **low reactivity**, making it valuable for industrial applications.
- Most of the graphite deposits are formed at **convergent plate boundaries**, where **organic-rich shales** and **limestones undergo metamorphism** due to intense heat and pressure. This process produces rocks like **marble**, **schist**, and **gneiss**, which contain tiny crystals and flakes of graphite. In some cases, graphite is formed through the metamorphism of coal seams, and this variety is known as **amorphous graphite**. Graphite is unique among non-metals because it is the **only non-metal that conducts electricity**.

- Natural graphite is widely used across various industries due to its valuable properties. It is essential in manufacturing **refractory materials** that retain strength at high temperatures, such as in **steelmaking**. Both natural and synthetic graphite are key components in **battery technologies**, including **lithium-ion batteries**, which require twice as much graphite as lithium carbonate. In **steelmaking**, it is used in **carbon raising** processes to strengthen steel. **Amorphous graphite** is used in **brake linings** for heavy vehicles, particularly as an alternative to asbestos. Graphite also serves as a specialty **lubricant** in extreme temperature conditions and is commonly found in **pencil leads** as a mixture with clay.

Production and Distribution of Graphite in India:

- India is rich in **graphite resources**, with notable occurrences across several states, including **Jammu and Kashmir**, **Gujarat**, **Jharkhand**, **Arunachal Pradesh**, **Karnataka**, **Kerala**, **Maharashtra**, **Tamil Nadu**, **Odisha**, **Chhattisgarh**, and **Rajasthan**.
- According to a 2013 report by the **Geological Survey of India (GSI)**, the state-wise distribution of **graphite resources** is led by **Arunachal Pradesh** with 43%, followed by **Jammu & Kashmir** with 37%, **Jharkhand** at 6%, **Tamil Nadu** at 5%, and **Odisha** at 3%.
- In terms of **production,** the majority is concentrated in **Tamil Nadu** (37%), **Jharkhand** (30%, particularly in the **Palamu district**), and **Odisha** (29%), making these states the key contributors to India's graphite output.

Conservation of Mineral Resources:

The conservation of mineral resources is one of the **most pressing issues facing the world today,** as minerals are **exhaustible resources**. Once extracted from the earth, these resources are permanently depleted. This finite nature of minerals is why mining is often referred to as a **"robber industry"**, as **it strips the earth of valuable non-renewable resources without the possibility of replenishment.**

The Accelerated Rate of Exploitation:

- The **advancement in mining technologies** has enabled the exploitation of minerals at a significantly faster pace. While this has contributed to economic growth, especially in countries like India, where a large number of minerals are exported to earn **much-needed foreign exchange**, it also raises concerns about sustainability. However, a better long-term strategy would be to **export goods manufactured from these minerals**, rather than exporting them in their raw form. This not only adds value but also helps conserve the resource by utilizing it more efficiently.

Improving Efficiency in Mining:

- **Improving mining efficiency** through advanced extraction and beneficiation technologies can reduce environmental impacts and minimize resource waste, ensuring more sustainable use of minerals.

Recycling of Minerals:

- Recycling **cyclic minerals** such as iron, aluminium, and copper can significantly reduce the need for fresh mining. Countries like Japan and Britain efficiently use scrap iron, conserving resources and cutting energy costs.

Substitution of Scarce Minerals:

- Another key strategy in conserving mineral resources is the **substitution of scarce minerals with more abundant and cheaper alternatives.** A notable example of this is the use of **aluminium in the electrical industry** to replace copper, which is becoming increasingly scarce. Aluminium is more abundant and offers a cost-effective alternative, helping to preserve dwindling copper reserves.

===== XXXXX =====

~~Northern~~ Chota Nagar Plataeu

→ Koderma Mica Mine largest in world

→ Coal field : Jharia, Raniganj,
Jhar. WB

→ Iron → Naomundi, Daltonganj, Mayurbanj
Jhark. Orissa

→ Sukinda chromite Mine } orissa

→ Uranium → Jaduguda } Jhar.
copper → Rakas

North East Mines

Mehlaya → coal field → Kopili
→ Uranium Mine → Mahadek basin
→ limestone → Mamwluh cave

Assam
↳ Oil & Gas
→ Disboi, Naharkatia,
↳ coal → Makum

A.P
↳ Nanphuk Namchik coal mine

Chattisgarh
Bailadila, Dallirajhara → Iron mine
Coal → Korba, Bilaspur, Jhilmili, Mahanadi

~~Chandrapur~~ → wardha river (Maharastra)

Southern belt

Kolar, Hatti-Gold Mine
Iron ore → Hospat mine
among
Baba Budan Hill
} Karnataka

Mica Mines : Nellore
Ramagiri Gold field
Coal among Neyelli region
} Andhra Pradesh

Some portion of Maha. & Goa has
Iron ore
ratanagiri, Mine & Bauxite among
western ghat

Gujrat
↳ Petroleum among Ankleswar, Mahesana, Kalol
↳ Bauxite among Nadid, Jauagarh

Rajasthan {
Khatri, Zavar Mine for
Pb, zinc & silver
Oil field among Mangla region